

Graph Stats

Graphs, Adjacency Lists, Degrees · Mentor-Led Lab (Student Edition)

Developed by TalenciaGlobal

Difficulty ★★ Intermediate

Problem

Read an undirected graph into an adjacency list and report its basic statistics: the number of vertices and edges, the degree of every vertex, and the maximum degree.

Implement the adjacency-list build and the degree computation from scratch.

What to Compute

- Vertices: the number of vertices N .
- Edges: the number of edges M .
- Degrees: the degree of each vertex $0..N-1$, in order.
- Max Degree: the largest degree.

Input / Output Format

Input: Line 1: N M . Next M lines: an undirected edge u v (vertices are $0..N-1$).

Output: Four lines: Vertices, Edges, Degrees, Max Degree.

Examples

Example 1

Input: 5 5 0 1 0 2 1 2 1 3 3 4

Output:

Vertices: 5
Edges: 5
Degrees: 2 3 2 2 1
Max Degree: 3

Explanation: Vertex 1 touches 0, 2 and 3 (degree 3, the maximum); the other degrees follow from the edge list.

Example 2

Input: 4 2 0 1 2 3

Output:

Vertices: 4

```
Edges: 2
Degrees: 1 1 1 1
Max Degree: 1
```

Explanation: Two disjoint edges give every vertex degree 1.

Constraints

- The graph is undirected; store each edge at both endpoints.
- Vertices are labelled 0 to N-1.
- Degree is the number of distinct neighbours.

Sample Run

Sample Input

```
5 5
0 1
0 2
1 2
1 3
3 4
```

Sample Output

```
Vertices: 5
Edges: 5
Degrees: 2 3 2 2 1
Max Degree: 3
```